Class 5: Data Viz with ggplot

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Background

There are lot's of ways to make plots in R. These include so-called "base R" (like the `plot()`) and add on packages like **ggplot2**.

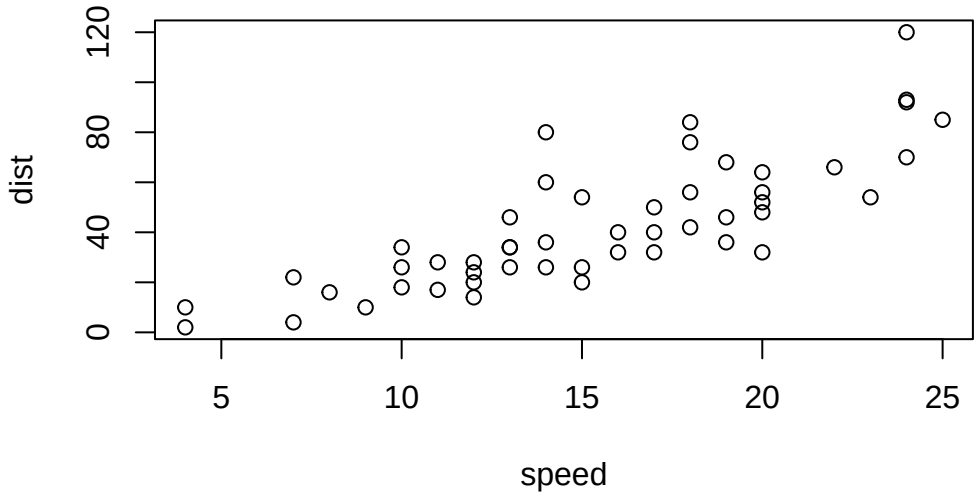
Let's make the same plot with these two graphics systems. We can use the inbuilt `cars` dataset:

```
head(cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

With "base R" we can simply:

```
plot(cars)
```



Now let's try ggplot. First I need to install the package using 'install.packages("ggplot2")'

> **N.B.** We never run an `install.packages()` in a code chunk otherwise we will re-install needlesely every time we render our document.

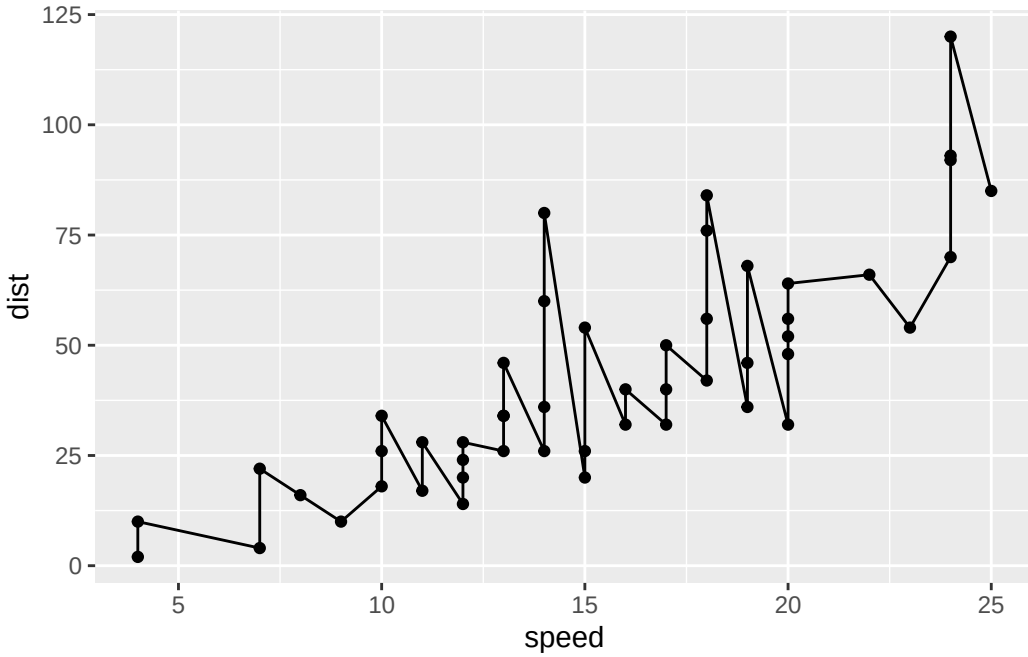
Every time we want to use an add-on package we need to load it up with a call to `library()`

```
library(ggplot2)
ggplot(cars)
```

Every gg plot needs at least 3 things:

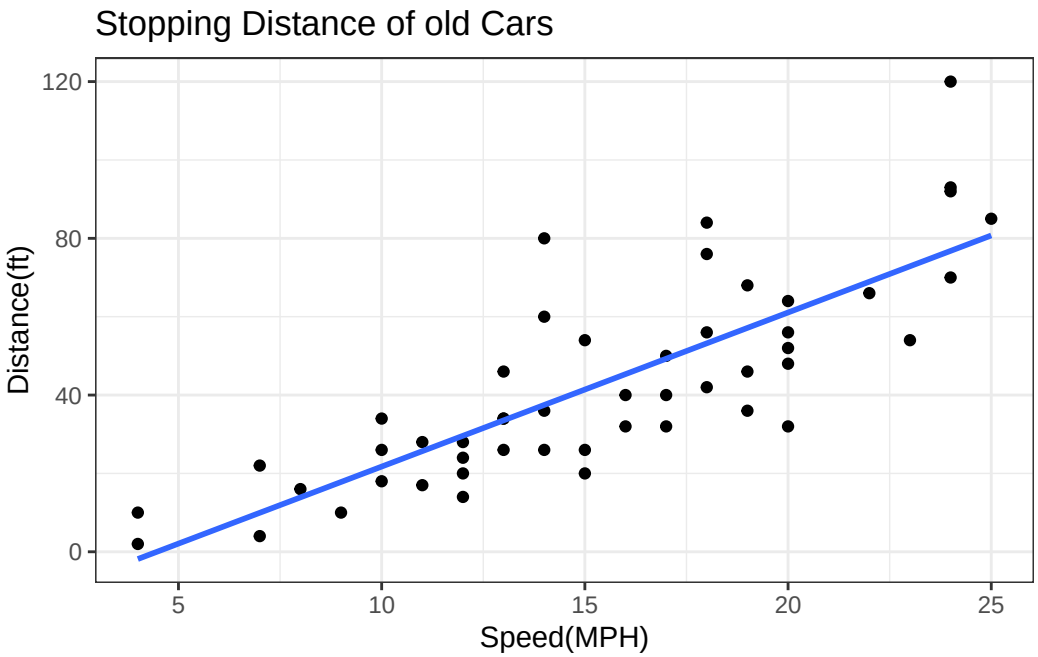
1. The **data** i.e. stuff to lot as a data.frame
2. The **aes** or aesthetics that map the data to the plot
3. The **geom_** or geometry i.e. the plot type such as points, lines etc.

```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  geom_line()
```




```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE)+
  labs(x="Speed(MPH)", y="Distance(ft)", title= "Stopping Distance of old Cars")+
  theme_bw()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Gene Expression Plot

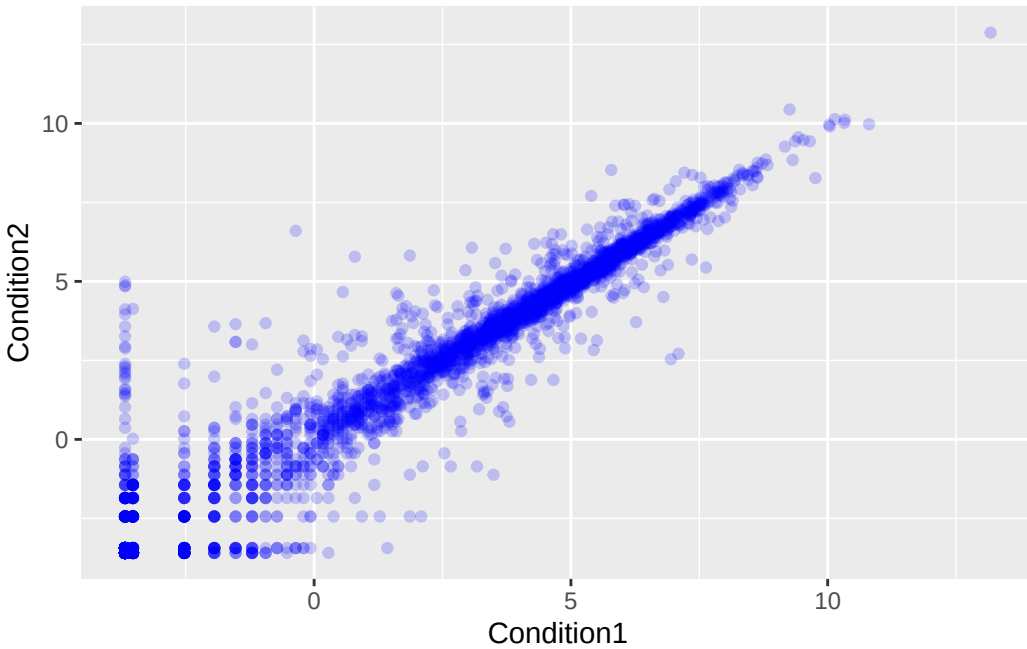
Read some data on the effects of GLP-1 inhibitor (drug) on gene expression values:

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

Version 1 plot - start simple by getting some ink on the page.

```
ggplot(genes) +
  aes(x=Condition1, y=Condition2) +
  geom_point(col="blue", alpha=0.2)
```

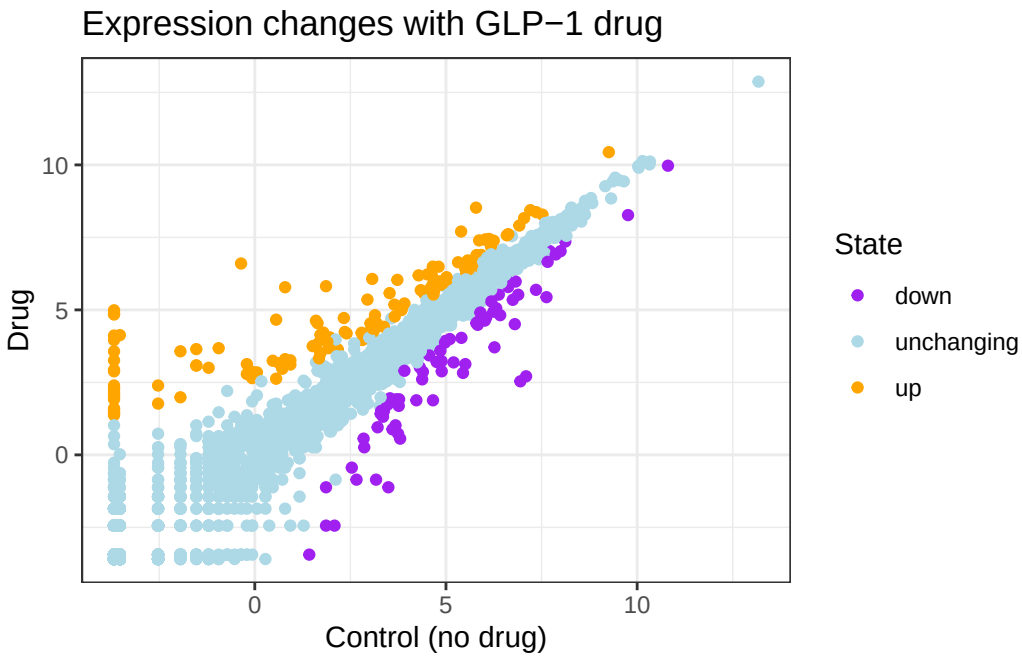


Let's color by `State` up, down or no change

```
table(genes$State)
```

```
      down unchanging         up
        72       4997        127
```

```
ggplot(genes) +
  aes(x=Condition1, y=Condition2, col=State) +
  geom_point() +
  scale_color_manual(values = c("purple", "lightblue","orange"))+
  theme_bw() +
  labs(x="Control (no drug)", y= "Drug", title = "Expression changes with GLP-1 drug")
```



Going further with gapmider

Here we explore the famous `gapmider` dataset with some custom plots.

```
# File location online
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv

gapminder <- read.delim(url)
head(gapminder)
```

```
      country continent year lifeExp      pop gdpPercap
1 Afghanistan      Asia 1952  28.801  8425333  779.4453
2 Afghanistan      Asia 1957  30.332  9240934  820.8530
3 Afghanistan      Asia 1962  31.997 10267083  853.1007
4 Afghanistan      Asia 1967  34.020 11537966  836.1971
5 Afghanistan      Asia 1972  36.088 13079460  739.9811
6 Afghanistan      Asia 1977  38.438 14880372  786.1134
```

Q. How many rows does this dataset have?

```r
nrow(gapminder)
```

```
[1] 1704
```

```r
nrow(genes)
```

```
[1] 5196
```

```r
ncol(genes)
```

```
[1] 4
```

How many different continents are in this dataset?

```r
table(gapminder$continent)
```

```
  Africa Americas     Asia   Europe  Oceania 
     624      300      396      360       24 
```

```r
table(genes$State)
```

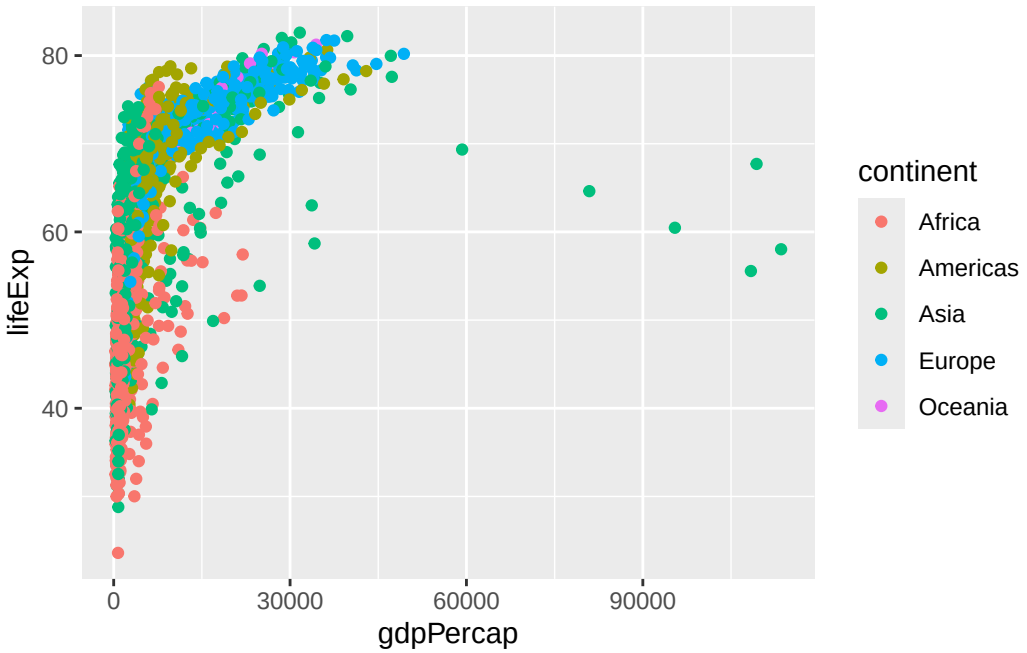
```
      down unchanging         up 
        72       4997        127 
```

```r
127/(72+4997+127)
```

```
[1] 0.02444188
```

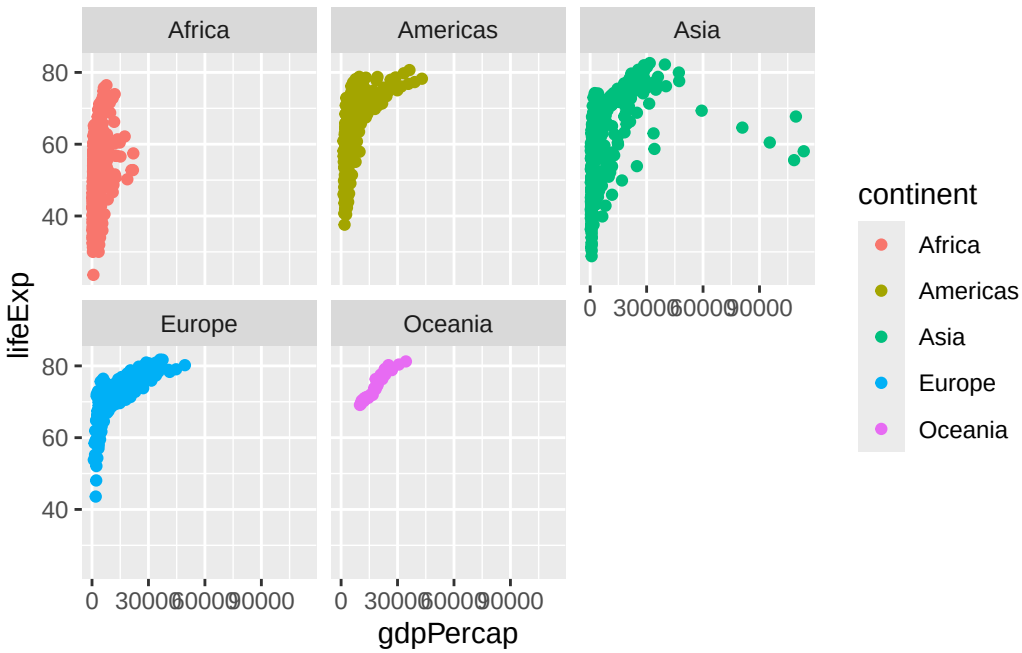
Version 1 plot gdpPercap vs LifeExp for all rows

```r
ggplot(gapminder) +
  aes(x=gdpPercap, y=lifeExp, col=continent) +
  geom_point()
```



I want to see a pot for each continent - in ggplot lingo this is called "faceting"

```
ggplot(gapminder) +
  aes(x=gdpPercap, y=lifeExp, col=continent) +
  geom_point()+
  facet_wrap(~continent)
```



First look at the dplyr package

Another add-on package with a function called `filter()` that we want to use.

```
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
filter(gapminder, year == 1952, country=="India")
```

```
  country continent year lifeExp       pop gdpPercap
1   India      Asia 1952  37.373 372000000  546.5657
```

```r
filter(gapminder, year == 2007 | year == 1977)
```

```
                    country continent year  lifeExp        pop  gdpPercap
1               Afghanistan      Asia 1977 38.43800   14880372   786.1134
2               Afghanistan      Asia 2007 43.82800   31889923   974.5803
3                   Albania    Europe 1977 68.93000    2509048  3533.0039
4                   Albania    Europe 2007 76.42300    3600523  5937.0295
5                   Algeria    Africa 1977 58.01400   17152804  4910.4168
6                   Algeria    Africa 2007 72.30100   33333216  6223.3675
7                    Angola    Africa 1977 39.48300    6162675  3008.6474
8                    Angola    Africa 2007 42.73100   12420476  4797.2313
9                 Argentina  Americas 1977 68.48100   26983828 10079.0267
10                Argentina  Americas 2007 75.32000   40301927 12779.3796
11                Australia   Oceania 1977 73.49000   14074100 18334.1975
12                Australia   Oceania 2007 81.23500   20434176 34435.3674
13                  Austria    Europe 1977 72.17000    7568430 19749.4223
14                  Austria    Europe 2007 79.82900    8199783 36126.4927
15                  Bahrain      Asia 1977 65.59300     297410 19340.1020
16                  Bahrain      Asia 2007 75.63500     708573 29796.0483
17               Bangladesh      Asia 1977 46.92300   80428306   659.8772
18               Bangladesh      Asia 2007 64.06200  150448339  1391.2538
19                  Belgium    Europe 1977 72.80000    9821800 19117.9745
20                  Belgium    Europe 2007 79.44100   10392226 33692.6051
21                    Benin    Africa 1977 49.19000    3168267  1029.1613
22                    Benin    Africa 2007 56.72800    8078314  1441.2849
23                  Bolivia  Americas 1977 50.02300    5079716  3548.0978
24                  Bolivia  Americas 2007 65.55400    9119152  3822.1371
25    Bosnia and Herzegovina    Europe 1977 69.86000    4086000  3528.4813
26    Bosnia and Herzegovina    Europe 2007 74.85200    4552198  7446.2988
27                 Botswana    Africa 1977 59.31900     781472  3214.8578
28                 Botswana    Africa 2007 50.72800    1639131 12569.8518
29                   Brazil  Americas 1977 61.48900  114313951  6660.1187
30                   Brazil  Americas 2007 72.39000  190010647  9065.8008
31                 Bulgaria    Europe 1977 70.81000    8797022  7612.2404
32                 Bulgaria    Europe 2007 73.00500    7322858 10680.7928
33             Burkina Faso    Africa 1977 46.13700    5889574   743.3870
34             Burkina Faso    Africa 2007 52.29500   14326203  1217.0330
35                  Burundi    Africa 1977 45.91000    3834415   556.1033
36                  Burundi    Africa 2007 49.58000    8390505   430.0707
37                 Cambodia      Asia 1977 31.22000    6978607   524.9722
38                 Cambodia      Asia 2007 59.72300   14131858  1713.7787
39                 Cameroon    Africa 1977 49.35500    7959865  1783.4329
```

40	Cameroon	Africa	2007	50.43000	17696293	2042.0952
41	Canada	Americas	1977	74.21000	23796400	22090.8831
42	Canada	Americas	2007	80.65300	33390141	36319.2350
43	Central African Republic	Africa	1977	46.77500	2167533	1109.3743
44	Central African Republic	Africa	2007	44.74100	4369038	706.0165
45	Chad	Africa	1977	47.38300	4388260	1133.9850
46	Chad	Africa	2007	50.65100	10238807	1704.0637
47	Chile	Americas	1977	67.05200	10599793	4756.7638
48	Chile	Americas	2007	78.55300	16284741	13171.6388
49	China	Asia	1977	63.96736	943455000	741.2375
50	China	Asia	2007	72.96100	1318683096	4959.1149
51	Colombia	Americas	1977	63.83700	25094412	3815.8079
52	Colombia	Americas	2007	72.88900	44227550	7006.5804
53	Comoros	Africa	1977	50.93900	304739	1172.6030
54	Comoros	Africa	2007	65.15200	710960	986.1479
55	Congo, Dem. Rep.	Africa	1977	47.80400	26480870	795.7573
56	Congo, Dem. Rep.	Africa	2007	46.46200	64606759	277.5519
57	Congo, Rep.	Africa	1977	55.62500	1536769	3259.1790
58	Congo, Rep.	Africa	2007	55.32200	3800610	3632.5578
59	Costa Rica	Americas	1977	70.75000	2108457	5926.8770
60	Costa Rica	Americas	2007	78.78200	4133884	9645.0614
61	Cote d'Ivoire	Africa	1977	52.37400	7459574	2517.7365
62	Cote d'Ivoire	Africa	2007	48.32800	18013409	1544.7501
63	Croatia	Europe	1977	70.64000	4318673	11305.3852
64	Croatia	Europe	2007	75.74800	4493312	14619.2227
65	Cuba	Americas	1977	72.64900	9537988	6380.4950
66	Cuba	Americas	2007	78.27300	11416987	8948.1029
67	Czech Republic	Europe	1977	70.71000	10161915	14800.1606
68	Czech Republic	Europe	2007	76.48600	10228744	22833.3085
69	Denmark	Europe	1977	74.69000	5088419	20422.9015
70	Denmark	Europe	2007	78.33200	5468120	35278.4187
71	Djibouti	Africa	1977	46.51900	228694	3081.7610
72	Djibouti	Africa	2007	54.79100	496374	2082.4816
73	Dominican Republic	Americas	1977	61.78800	5302800	2681.9889
74	Dominican Republic	Americas	2007	72.23500	9319622	6025.3748
75	Ecuador	Americas	1977	61.31000	7278866	6679.6233
76	Ecuador	Americas	2007	74.99400	13755680	6873.2623
77	Egypt	Africa	1977	53.31900	38783863	2785.4936
78	Egypt	Africa	2007	71.33800	80264543	5581.1810
79	El Salvador	Americas	1977	56.69600	4282586	5138.9224
80	El Salvador	Americas	2007	71.87800	6939688	5728.3535
81	Equatorial Guinea	Africa	1977	42.02400	192675	958.5668
82	Equatorial Guinea	Africa	2007	51.57900	551201	12154.0897

83	Eritrea	Africa	1977	44.53500	2512642	505.7538
84	Eritrea	Africa	2007	58.04000	4906585	641.3695
85	Ethiopia	Africa	1977	44.51000	34617799	556.8084
86	Ethiopia	Africa	2007	52.94700	76511887	690.8056
87	Finland	Europe	1977	72.52000	4738902	15605.4228
88	Finland	Europe	2007	79.31300	5238460	33207.0844
89	France	Europe	1977	73.83000	53165019	18292.6351
90	France	Europe	2007	80.65700	61083916	30470.0167
91	Gabon	Africa	1977	52.79000	706367	21745.5733
92	Gabon	Africa	2007	56.73500	1454867	13206.4845
93	Gambia	Africa	1977	41.84200	608274	884.7553
94	Gambia	Africa	2007	59.44800	1688359	752.7497
95	Germany	Europe	1977	72.50000	78160773	20512.9212
96	Germany	Europe	2007	79.40600	82400996	32170.3744
97	Ghana	Africa	1977	51.75600	10538093	993.2240
98	Ghana	Africa	2007	60.02200	22873338	1327.6089
99	Greece	Europe	1977	73.68000	9308479	14195.5243
100	Greece	Europe	2007	79.48300	10706290	27538.4119
101	Guatemala	Americas	1977	56.02900	5703430	4879.9927
102	Guatemala	Americas	2007	70.25900	12572928	5186.0500
103	Guinea	Africa	1977	40.76200	4227026	874.6859
104	Guinea	Africa	2007	56.00700	9947814	942.6542
105	Guinea-Bissau	Africa	1977	37.46500	745228	764.7260
106	Guinea-Bissau	Africa	2007	46.38800	1472041	579.2317
107	Haiti	Americas	1977	49.92300	4908554	1874.2989
108	Haiti	Americas	2007	60.91600	8502814	1201.6372
109	Honduras	Americas	1977	57.40200	3055235	3203.2081
110	Honduras	Americas	2007	70.19800	7483763	3548.3308
111	Hong Kong, China	Asia	1977	73.60000	4583700	11186.1413
112	Hong Kong, China	Asia	2007	82.20800	6980412	39724.9787
113	Hungary	Europe	1977	69.95000	10637171	11674.8374
114	Hungary	Europe	2007	73.33800	9956108	18008.9444
115	Iceland	Europe	1977	76.11000	221823	19654.9625
116	Iceland	Europe	2007	81.75700	301931	36180.7892
117	India	Asia	1977	54.20800	634000000	813.3373
118	India	Asia	2007	64.69800	1110396331	2452.2104
119	Indonesia	Asia	1977	52.70200	136725000	1382.7021
120	Indonesia	Asia	2007	70.65000	223547000	3540.6516
121	Iran	Asia	1977	57.70200	35480679	11888.5951
122	Iran	Asia	2007	70.96400	69453570	11605.7145
123	Iraq	Asia	1977	60.41300	11882916	14688.2351
124	Iraq	Asia	2007	59.54500	27499638	4471.0619
125	Ireland	Europe	1977	72.03000	3271900	11150.9811

126	Ireland	Europe	2007	78.88500	4109086	40675.9964
127	Israel	Asia	1977	73.06000	3495918	13306.6192
128	Israel	Asia	2007	80.74500	6426679	25523.2771
129	Italy	Europe	1977	73.48000	56059245	14255.9847
130	Italy	Europe	2007	80.54600	58147733	28569.7197
131	Jamaica	Americas	1977	70.11000	2156814	6650.1956
132	Jamaica	Americas	2007	72.56700	2780132	7320.8803
133	Japan	Asia	1977	75.38000	113872473	16610.3770
134	Japan	Asia	2007	82.60300	127467972	31656.0681
135	Jordan	Asia	1977	61.13400	1937652	2852.3516
136	Jordan	Asia	2007	72.53500	6053193	4519.4612
137	Kenya	Africa	1977	56.15500	14500404	1267.6132
138	Kenya	Africa	2007	54.11000	35610177	1463.2493
139	Korea, Dem. Rep.	Asia	1977	67.15900	16325320	4106.3012
140	Korea, Dem. Rep.	Asia	2007	67.29700	23301725	1593.0655
141	Korea, Rep.	Asia	1977	64.76600	36436000	4657.2210
142	Korea, Rep.	Asia	2007	78.62300	49044790	23348.1397
143	Kuwait	Asia	1977	69.34300	1140357	59265.4771
144	Kuwait	Asia	2007	77.58800	2505559	47306.9898
145	Lebanon	Asia	1977	66.09900	3115787	8659.6968
146	Lebanon	Asia	2007	71.99300	3921278	10461.0587
147	Lesotho	Africa	1977	52.20800	1251524	745.3695
148	Lesotho	Africa	2007	42.59200	2012649	1569.3314
149	Liberia	Africa	1977	43.76400	1703617	640.3224
150	Liberia	Africa	2007	45.67800	3193942	414.5073
151	Libya	Africa	1977	57.44200	2721783	21951.2118
152	Libya	Africa	2007	73.95200	6036914	12057.4993
153	Madagascar	Africa	1977	46.88100	8007166	1544.2286
154	Madagascar	Africa	2007	59.44300	19167654	1044.7701
155	Malawi	Africa	1977	43.76700	5637246	663.2237
156	Malawi	Africa	2007	48.30300	13327079	759.3499
157	Malaysia	Asia	1977	65.25600	12845381	3827.9216
158	Malaysia	Asia	2007	74.24100	24821286	12451.6558
159	Mali	Africa	1977	41.71400	6491649	686.3953
160	Mali	Africa	2007	54.46700	12031795	1042.5816
161	Mauritania	Africa	1977	50.85200	1456688	1497.4922
162	Mauritania	Africa	2007	64.16400	3270065	1803.1515
163	Mauritius	Africa	1977	64.93000	913025	3710.9830
164	Mauritius	Africa	2007	72.80100	1250882	10956.9911
165	Mexico	Americas	1977	65.03200	63759976	7674.9291
166	Mexico	Americas	2007	76.19500	108700891	11977.5750
167	Mongolia	Asia	1977	55.49100	1528000	1647.5117
168	Mongolia	Asia	2007	66.80300	2874127	3095.7723

169	Montenegro	Europe	1977	73.06600	560073	9595.9299
170	Montenegro	Europe	2007	74.54300	684736	9253.8961
171	Morocco	Africa	1977	55.73000	18396941	2370.6200
172	Morocco	Africa	2007	71.16400	33757175	3820.1752
173	Mozambique	Africa	1977	42.49500	11127868	502.3197
174	Mozambique	Africa	2007	42.08200	19951656	823.6856
175	Myanmar	Asia	1977	56.05900	31528087	371.0000
176	Myanmar	Asia	2007	62.06900	47761980	944.0000
177	Namibia	Africa	1977	56.43700	977026	3876.4860
178	Namibia	Africa	2007	52.90600	2055080	4811.0604
179	Nepal	Asia	1977	46.74800	13933198	694.1124
180	Nepal	Asia	2007	63.78500	28901790	1091.3598
181	Netherlands	Europe	1977	75.24000	13852989	21209.0592
182	Netherlands	Europe	2007	79.76200	16570613	36797.9333
183	New Zealand	Oceania	1977	72.22000	3164900	16233.7177
184	New Zealand	Oceania	2007	80.20400	4115771	25185.0091
185	Nicaragua	Americas	1977	57.47000	2554598	5486.3711
186	Nicaragua	Americas	2007	72.89900	5675356	2749.3210
187	Niger	Africa	1977	41.29100	5682086	808.8971
188	Niger	Africa	2007	56.86700	12894865	619.6769
189	Nigeria	Africa	1977	44.51400	62209173	1981.9518
190	Nigeria	Africa	2007	46.85900	135031164	2013.9773
191	Norway	Europe	1977	75.37000	4043205	23311.3494
192	Norway	Europe	2007	80.19600	4627926	49357.1902
193	Oman	Asia	1977	57.36700	1004533	11848.3439
194	Oman	Asia	2007	75.64000	3204897	22316.1929
195	Pakistan	Asia	1977	54.04300	78152686	1175.9212
196	Pakistan	Asia	2007	65.48300	169270617	2605.9476
197	Panama	Americas	1977	68.68100	1839782	5351.9121
198	Panama	Americas	2007	75.53700	3242173	9809.1856
199	Paraguay	Americas	1977	66.35300	2984494	3248.3733
200	Paraguay	Americas	2007	71.75200	6667147	4172.8385
201	Peru	Americas	1977	58.44700	15990099	6281.2909
202	Peru	Americas	2007	71.42100	28674757	7408.9056
203	Philippines	Asia	1977	60.06000	46850962	2373.2043
204	Philippines	Asia	2007	71.68800	91077287	3190.4810
205	Poland	Europe	1977	70.67000	34621254	9508.1415
206	Poland	Europe	2007	75.56300	38518241	15389.9247
207	Portugal	Europe	1977	70.41000	9662600	10172.4857
208	Portugal	Europe	2007	78.09800	10642836	20509.6478
209	Puerto Rico	Americas	1977	73.44000	3080828	9770.5249
210	Puerto Rico	Americas	2007	78.74600	3942491	19328.7090
211	Reunion	Africa	1977	67.06400	492095	4319.8041

212	Reunion	Africa	2007	76.44200	798094	7670.1226
213	Romania	Europe	1977	69.46000	21658597	9356.3972
214	Romania	Europe	2007	72.47600	22276056	10808.4756
215	Rwanda	Africa	1977	45.00000	4657072	670.0806
216	Rwanda	Africa	2007	46.24200	8860588	863.0885
217	Sao Tome and Principe	Africa	1977	58.55000	86796	1737.5617
218	Sao Tome and Principe	Africa	2007	65.52800	199579	1598.4351
219	Saudi Arabia	Asia	1977	58.69000	8128505	34167.7626
220	Saudi Arabia	Asia	2007	72.77700	27601038	21654.8319
221	Senegal	Africa	1977	48.87900	5260855	1561.7691
222	Senegal	Africa	2007	63.06200	12267493	1712.4721
223	Serbia	Europe	1977	70.30000	8686367	12980.6696
224	Serbia	Europe	2007	74.00200	10150265	9786.5347
225	Sierra Leone	Africa	1977	36.78800	3140897	1348.2852
226	Sierra Leone	Africa	2007	42.56800	6144562	862.5408
227	Singapore	Asia	1977	70.79500	2325300	11210.0895
228	Singapore	Asia	2007	79.97200	4553009	47143.1796
229	Slovak Republic	Europe	1977	70.45000	4827803	10922.6640
230	Slovak Republic	Europe	2007	74.66300	5447502	18678.3144
231	Slovenia	Europe	1977	70.97000	1746919	15277.0302
232	Slovenia	Europe	2007	77.92600	2009245	25768.2576
233	Somalia	Africa	1977	41.97400	4353666	1450.9925
234	Somalia	Africa	2007	48.15900	9118773	926.1411
235	South Africa	Africa	1977	55.52700	27129932	8028.6514
236	South Africa	Africa	2007	49.33900	43997828	9269.6578
237	Spain	Europe	1977	74.39000	36439000	13236.9212
238	Spain	Europe	2007	80.94100	40448191	28821.0637
239	Sri Lanka	Asia	1977	65.94900	14116836	1348.7757
240	Sri Lanka	Asia	2007	72.39600	20378239	3970.0954
241	Sudan	Africa	1977	47.80000	17104986	2202.9884
242	Sudan	Africa	2007	58.55600	42292929	2602.3950
243	Swaziland	Africa	1977	52.53700	551425	3781.4106
244	Swaziland	Africa	2007	39.61300	1133066	4513.4806
245	Sweden	Europe	1977	75.44000	8251648	18855.7252
246	Sweden	Europe	2007	80.88400	9031088	33859.7484
247	Switzerland	Europe	1977	75.39000	6316424	26982.2905
248	Switzerland	Europe	2007	81.70100	7554661	37506.4191
249	Syria	Asia	1977	61.19500	7932503	3195.4846
250	Syria	Asia	2007	74.14300	19314747	4184.5481
251	Taiwan	Asia	1977	70.59000	16785196	5596.5198
252	Taiwan	Asia	2007	78.40000	23174294	28718.2768
253	Tanzania	Africa	1977	49.91900	17129565	962.4923
254	Tanzania	Africa	2007	52.51700	38139640	1107.4822

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255                Thailand      Asia 1977 62.49400   44148285  1961.2246
256                Thailand      Asia 2007 70.61600   65068149  7458.3963
257                    Togo    Africa 1977 52.88700    2308582  1532.7770
258                    Togo    Africa 2007 58.42000    5701579   882.9699
259      Trinidad and Tobago  Americas 1977 68.30000    1039009  7899.5542
260      Trinidad and Tobago  Americas 2007 69.81900    1056608 18008.5092
261                  Tunisia    Africa 1977 59.83700    6005061  3120.8768
262                  Tunisia    Africa 2007 73.92300   10276158  7092.9230
263                   Turkey    Europe 1977 59.50700   42404033  4269.1223
264                   Turkey    Europe 2007 71.77700   71158647  8458.2764
265                   Uganda    Africa 1977 50.35000   11457758   843.7331
266                   Uganda    Africa 2007 51.54200   29170398  1056.3801
267           United Kingdom    Europe 1977 72.76000   56179000 17428.7485
268           United Kingdom    Europe 2007 79.42500   60776238 33203.2613
269            United States  Americas 1977 73.38000  220239000 24072.6321
270            United States  Americas 2007 78.24200  301139947 42951.6531
271                  Uruguay  Americas 1977 69.48100    2873520  6504.3397
272                  Uruguay  Americas 2007 76.38400    3447496 10611.4630
273                Venezuela  Americas 1977 67.45600   13503563 13143.9510
274                Venezuela  Americas 2007 73.74700   26084662 11415.8057
275                  Vietnam      Asia 1977 55.76400   50533506   713.5371
276                  Vietnam      Asia 2007 74.24900   85262356  2441.5764
277       West Bank and Gaza      Asia 1977 60.76500    1261091  3682.8315
278       West Bank and Gaza      Asia 2007 73.42200    4018332  3025.3498
279              Yemen, Rep.      Asia 1977 44.17500    8403990  1829.7652
280              Yemen, Rep.      Asia 2007 62.69800   22211743  2280.7699
281                   Zambia    Africa 1977 51.38600    5216550  1588.6883
282                   Zambia    Africa 2007 42.38400   11746035  1271.2116
283                 Zimbabwe    Africa 1977 57.67400    6642107   685.5877
284                 Zimbabwe    Africa 2007 43.48700   12311143   469.7093
```

```r
ggplot(gapminder)+
  aes(x=gdpPercap, y=lifeExp, col=continent)+
  geom_point()+
  filter(gapminder, year == 2007 | year == 1977)+
  facet_wrap(~year)
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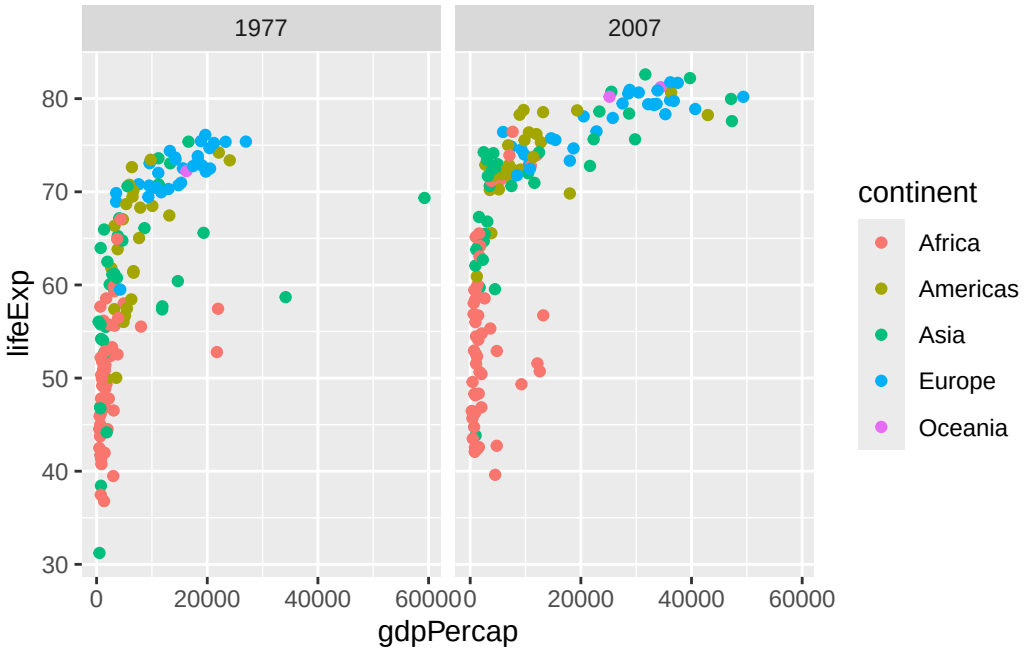
1977
2007
lifeExp
80
70
60
50
40
30
0
20000
40000
60000
0
20000
40000
60000
gdpPercap
continent
Africa
Americas
Asia
Europe
Oceania